

Directory Structure (SDE Question) - Part A

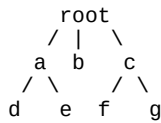
Question 1

You are given a directory structure represented as a tree, where each node represents a directory that can have any number of child directories.

- Each child directory will have a unique name.
- The second line of the input will contain the root name and its child directories.

Based on the given input, you have to construct the directory structure and implement three functions: **countDescendants**, **cutPaste** and **copyPaste**.

Example Tree:



Function Details

1. countDescendants(path)

- Takes one argument: directory path.
- Returns the number of descendants of the directory.

Examples:

```
countDescendants("root")      => 7
countDescendants("root/a")    => 2
countDescendants("root/b")    => 0
countDescendants("root/c")    => 2
countDescendants("root/a/d")  => 0
countDescendants("root/a/e")  => 0
```

2. cutPaste(src, dest)

- It takes two arguments: source directory and destination directory.
- Cuts the source directory from its parent and pastes it inside the destination directory.

Example:

```
cutPaste("root/a", "root/c")
```

The source directory **a** (including its children) is moved under **c**.

3. copyPaste(src, dest)

- It takes two arguments: source directory and destination directory.
- Copies the source directory and its entire subtree, and pastes the copy into the destination directory.

Example:

```
copyPaste("root/a", "root/c")
```

The directory **a** and its children are copied under **c**; the original **a** remains under **root**.